Code No: 135AF **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, March - 2024
DESIGN AND ANALYSIS OF ALGORITHMS
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

Useful Papers March-2024

PART – A

(25 Marks)

1.a)	What is time complexity?	[2]
b)	What are randomized algorithms?	[3]
c)	What are biconnected components?	[2]
d)	Write down the algorithm for union().	[3]
e)	Distinguish between backtracking and greedy approach.	[2]
f)	What are the applications of single source shortest path?	[3]
g)	What is chained matrix multiplication?	[2]
h)	What is time and space complexity of all pairs shortest path using dynamic programming approach?	[3]
i)	What is NP hard problem?	[2]
j)	What is meant by state space tree?	[3]

PART – B

(50 Marks)

2. What is the basic idea of Quick sort algorithm? Write an algorithm for quick sort and derive its time complexity. Also perform quicksort on the following data: [10]
3,2,8,9,11,42,27,5,7

OR

3. Explain strassen's matrix multiplication? Show how time complexity of this algorithm is reduced through divide and conquer method. [10]

4. Write an algorithm and explain how to find Hamiltonian cycles through backtracking approach. [10]

OR

5. What is n-queens problem? Write an algorithm and find the solution to it using backtracking approach. [10]

6. What is minimum cost spanning tree? Explain prim's and krushkal's algorithm to find minimum cost spanning tree with an illustrative example. [10]

OR

7. State and explain knapsack problem. Give solution to it using greedy approach. [10]

8. What are optimal binary search trees? Write down algorithm for it and derive its time complexity. [10]

OR

9. Define travelling salesperson problem. Give solution to it using backtracking approach. Write down algorithm for it and illustrate with an example. [10]

10. Give FIFO Branch and Bound solution to 0/1 Knapsack problem. [10]

OR

11.a) State and explain cook's theorem.

b) Explain about NP complete problems. [5+5]

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